

Question 1

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Which one of the following is asynchronous in message communication?

- ☐ a. Direct message
- ☐ b. Non-block receive
- ☐ c. Blocking send
- ☐ d. Blocking receive

The correct answer is B. Non-blocking receive.

In the given options, the only choice that represents an asynchronous message in communication is B. Non-blocking receive.

Asynchronous communication refers to a method where participants do not require immediate or real-time interaction. In asynchronous messaging, participants can send and receive messages independently and at their own convenience, without the need for immediate responses.

In the given options: A. Direct Message: This does not necessarily indicate whether it is synchronous or asynchronous, as it depends on the specific implementation. It could be either synchronous or asynchronous. B. Non-blocking receive: This represents an asynchronous message. It indicates that the receiver can retrieve a message without being blocked or waiting for an immediate response from the sender. C. Blocking Send: This represents a synchronous message, where the sender is expected to wait for a response before proceeding. D. Blocking receive: This also represents a synchronous message, where the receiver is expected to wait for a message before proceeding.

Therefore, among the options provided, B. Non-blocking receive is the only one that represents an asynchronous message in communication.

Question 2

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What is the time complexity order of binary searching algorithm?

- ☐ a. $O(n)$
- ☐ b. $O(\log_2 n)$
- ☐ c. $O(n^3)$
- ☐ d. $O(1)$

Answer B;

The time complexity of the binary search algorithm is $O(\log n)$, where "n" represents the number of elements in the sorted array being searched.

Binary search follows a divide-and-conquer strategy to locate a specific element within a sorted array. It starts by comparing the target element with the middle element of the array. If they match, the search is successful. If the target element is smaller, the algorithm continues searching in the left half of the

array; otherwise, it searches in the right half. This process of dividing the search space in half is repeated until the target element is found or the search space is exhausted.

Each iteration of the binary search reduces the search space by half. As a result, the number of remaining elements to be searched becomes halved with each iteration. Therefore, the algorithm's time complexity is logarithmic, specifically base 2.

In terms of Big O notation, $O(\log n)$ represents an efficient time complexity, as the number of operations required to find the target element grows at a slower rate compared to linear search ($O(n)$), where elements are searched one by one.

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Question 3
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Which one of the following is not true about fan trap in Entity Relationship Diagram?

- ☐ a. It occurs when there is no pathway between related entities
- ☐ b. Highly likely exit when more than two one-to-many relationships fan out from an entity
- ☐ c. When there is ambiguous pathway between some entity occurrences in related entity types
- ☐ d. It can be resolved by readjusting the model so that no one-to-many relationships fanning out from an entity

Answer A;

The E-R model can result problems due to limitations in the way the entities are related in the relational databases. These problems are called connection traps. These problems often occur due to a misinterpretation of the meaning of certain relationships.

Two main types of connection traps are called fan traps and chasm traps.

Fan Trap. It occurs when a model represents a relationship between entity types, but pathway between certain entity occurrences is ambiguous.

A fan trap occurs when one to many relationships fan out from a single entity.

For example: Consider a [database](#) of Department, Site and Staff, where one site can contain number of department, but a department is situated only at a single site. There are multiple staff members working at a single site and a staff member can work from a single site. The above case is represented in e-r diagram shown. The problem of above e-r diagram is that, which staff works in a particular department remain answered. The solution is to restructure the original E-R model to represent the correct association as shown.

Chasm Trap. It occurs when a model suggests the existence of a relationship between entity types, but pathway does not exist between certain entity occurrences.

It occurs where there is a relationship with partial participation, which forms part of the pathway between entities that are related.

For example: Let us consider a [database](#) where, a single branch is allocated many staff who handles the management of properties for rent. Not all staff members handle the

property and not all property is managed by a member of staff. The above case is represented in the e-r diagram.

see more visit: <https://ecomputernotes.com/fundamental/what-is-a-database/problems-with-e-r-model>

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Question 4
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Which concept refers to extending specialized classes from generalized classes in object-oriented programming?

- ☐ a. Data field Encapsulation
- ☐ b. Inheritance
- ☐ c. Overloading
- ☐ d. Overriding

B. Inheritance

In object-oriented programming (OOP), inheritance, overloading, overriding, and data field encapsulation are fundamental concepts that help in creating modular and reusable code. Here are their definitions:

1. **Inheritance:** Inheritance is a mechanism in OOP that allows a class (called a subclass or specialized or derived class) to inherit properties and behaviors from another class (called a superclass or generalised or base class). The subclass can reuse and extend the features of the superclass, which promotes code reuse and supports the concept of "is-a" relationship.

By inheriting from a superclass, the subclass automatically gains access to the superclass's methods and attributes. It can also add its own unique methods and attributes. Inheritance enables the creation of hierarchical relationships among classes, where more specialized classes inherit from more general classes.

2. **Overloading:** Overloading is a feature in OOP that allows multiple methods or functions with the same name but different parameters to coexist in a class. The methods or functions can have the same name but must have different parameter lists (different types, different number of parameters, or both).

Overloading enables developers to define multiple methods or functions that perform similar tasks but operate on different data types or with different parameter combinations. The appropriate method or function to execute is determined at compile-time or runtime based on the method or function signature and the arguments passed.

3. **Overriding:** Overriding is a concept in OOP that allows a subclass to provide a different implementation of a method that is already defined in its superclass. When a method in the

subclass has the same name, return type, and parameters as a method in the superclass, it overrides the superclass's method.

By overriding a method, the subclass can customize the behavior of the inherited method to suit its specific requirements. This enables polymorphism, where objects of different classes can be treated as instances of a common superclass but exhibit different behaviors based on their specific implementations.

4. **Data Field Encapsulation:** Data field encapsulation, also known as data encapsulation or data hiding, is a principle in OOP that emphasizes the bundling of data and methods that operate on that data within a class. It involves making the internal state of an object private and providing controlled access to that state through public methods (getters and setters).

Encapsulation helps in achieving data abstraction and information hiding. It allows for the protection and control of data integrity by preventing direct access to the internal data fields. Access to the data is provided through well-defined methods, which can enforce validation, implement business logic, or maintain consistency.

Encapsulation promotes modular code, as it allows objects to interact with each other through well-defined interfaces while hiding their internal details. It enhances code maintainability, flexibility, and security.

In summary, inheritance enables code reuse and supports hierarchical relationships, overloading allows multiple methods with the same name but different parameters, overriding allows a subclass to provide a different implementation of a method in the superclass, and data field encapsulation promotes data protection and controlled access through methods.

Question 5

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Which one of the following registers hold the address of the next pair of instructions to be fetched from memory?

- ☐ a. Memory address register
- ☐ b. Instruction buffer register
- ☐ c. Memory buffer register
- ☐ d. Program counter

Answer D;

In Computer Organization and Architecture (COA), several registers play crucial roles in the functioning of a computer system. Here are brief explanations of the memory address register, instruction buffer register, memory buffer register, and program counter:

1. **Memory Address Register (MAR):** The Memory Address Register (MAR) is a special register that holds the memory address of the data or instruction being accessed or stored in the main memory. It is used in the memory fetch or store operations. When a memory read or write

operation is initiated, the MAR holds the address where the data or instruction is located or where it will be stored.

2. **Instruction Buffer Register (IBR):** The Instruction Buffer Register (IBR), also known as the Instruction Register (IR), is a register that holds the current instruction being executed by the processor. It serves as a temporary storage location for the fetched instruction from the memory. The instruction stored in the IBR is then decoded and executed by the processor's control unit.
3. **Memory Buffer Register (MBR):** The Memory Buffer Register (MBR) is a register that holds the data being transferred between the processor and the memory. When data is read from memory, it is temporarily stored in the MBR before being transferred to other registers or the arithmetic logic unit (ALU) for further processing. Similarly, when data is written to memory, it is first stored in the MBR before being transferred to the memory location specified by the MAR.
4. **Program Counter (PC):** The Program Counter (PC) is a register that keeps track of the memory address of the next instruction to be fetched and executed in a program. It holds the address of the instruction in the memory that the processor should fetch next. After each instruction is executed, the PC is updated to point to the next instruction in sequence. It plays a crucial role in the program flow control, ensuring that instructions are executed in the correct order.

In summary, the Memory Address Register (MAR) holds the memory address of the data or instruction, the Instruction Buffer Register (IBR) temporarily stores the current instruction being executed, the Memory Buffer Register (MBR) holds the data being transferred between the processor and memory, and the Program Counter (PC) keeps track of the memory address of the next instruction to be fetched and executed. These registers are essential for the smooth operation and control of a computer system.

Question 6

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Size of large data files can be reduced to save storage disk space. Which algorithm is appropriate to reduce size of large files?

- ☐ a. Huffman encoding algorithm
- ☐ b. Merge sort algorithm
- ☐ c. Heap sort algorithm
- ☐ d. Prim's algorithm

Answer A;

1. **Huffman Coding:** it is used for both data compression and decompression efficiently without data loss and also it is an examples of Greedy algorithms.

Huffman coding is a data compression algorithm used to efficiently encode and compress data. It uses variable-length codes, assigning shorter codes to more frequently occurring characters or symbols, and longer codes to less frequent ones. This results in a compression scheme that optimally utilizes the available space. Huffman coding finds applications in file compression, image compression, video compression, and other data compression scenarios.

2. **Merge Sort:** Merge sort is a comparison-based sorting algorithm that operates by recursively dividing the input array into smaller subarrays, sorting them, and then merging them back together. It follows a divide-and-conquer strategy and guarantees a time complexity of $O(n \log n)$, making it efficient for large datasets. Merge sort is widely used in various applications where sorting is required, such as database systems, external sorting, and parallel computing.
3. **Heap Sort:** Heap sort is another comparison-based sorting algorithm that uses a binary heap data structure. It starts by constructing a max-heap or min-heap (depending on the desired order) from the input array. Then, it repeatedly extracts the maximum or minimum element from the heap and places it at the end of the sorted array. Heap sort has a time complexity of $O(n \log n)$ and is often used in situations where in-place sorting is required and the input size is moderate.
4. **Prim's Algorithm:** Prim's algorithm is a greedy algorithm used for finding the minimum spanning tree (MST) of a weighted undirected graph. Starting from an arbitrary vertex, Prim's algorithm progressively adds the edge with the minimum weight that connects a vertex in the MST to a vertex outside the MST. This process continues until all vertices are included in the MST. Prim's algorithm finds applications in network design, clustering, and data analysis, where the objective is to connect nodes or vertices in an efficient and cost-effective manner.

In summary, Huffman coding is used for data compression, merge sort is used for efficient sorting, heap sort is used for in-place sorting, and Prim's algorithm is used for finding minimum spanning trees in graphs. Each algorithm has its own unique applications, helping to solve specific problems efficiently in various domains.

Question 7

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_____ is set of attributes that can be seen by a program and can have direct impact on the logical execution of a program.

- ☐ a. Organization
- ☐ b. Arrangement
- ☐ c. Architecture
- ☐ d. Flow control

Answer C: Architecture; **Computer architecture** refers to those attributes of a system visible to a programmer or, put another way, those attributes that have a direct impact on the logical execution of a program.

Computer Architecture	Computer Organization
Computer Architecture is concerned with the way hardware components are connected together to form a computer system.	Computer Organization is concerned with the structure and behaviour of a computer system as seen by the user.
It acts as the interface between hardware and software.	It deals with the components of a connection in a system.
Computer Architecture helps us to understand the functionalities of a system.	Computer Organization tells us how exactly all the units in the system are arranged and interconnected.
A programmer can view architecture in terms of instructions, addressing modes and registers.	Whereas Organization expresses the realization of architecture.
While designing a computer system architecture is considered first.	An organization is done on the basis of architecture.
Computer Architecture deals with high-level design issues.	Computer Organization deals with low-level design issues.
Architecture involves Logic (Instruction sets, Addressing modes, Data types, Cache optimization)	Organization involves Physical Components (Circuit design, Adders, Signals, Peripherals)

Question 8

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Which one of the following is true about frames in HTML?

- ☐ a. Frames allow parts of the page to remain stationary while other parts scroll
- ☐ b. Load on the server is not affected, if there are a large number of frames in a page
- ☐ c. All browsers support frames
- ☐ d. Frames are not difficult to handle for search engines

Answer A;

HTML frames are used to divide your browser window into multiple sections where each section can load a separate HTML document. A collection of frames in the browser window is known as a frameset. The window is divided into frames in a similar way the tables are organized: into rows and columns.

We can create stationary and scroll part on the webpage using frames by utilizing CSS.

Advantages:

- It allows the user to view multiple documents within a single Web page.

- It load pages from different servers in a single frameset.
- The older browsers that do not support frames can be addressed using the tag.

Disadvantages: Due to some of its disadvantage it is rarely used in web browser.

- Frames can make the production of website complicated.
- A user is unable to bookmark any of the Web pages viewed within a frame.
- The browser's back button might not work as the user hopes.
- The use of too many frames can put a high workload on the server.
- Many old web browser doesn't support frames.

Note: <frame> tag is not supported in HTML5 (but <iframe> is supported).

Supported Browser: The browser supported by <frame> tag are listed below:

- Google Chrome
- Internet Explorer
- Firefox
- Opera
- Safari

Each frame in a webpage requires a separate request to the server to retrieve its content. As the number of frames increases, the server needs to handle more requests simultaneously, which can lead to increased server load and potentially impact its performance.

Moreover, frames typically load their content independently, which means that each frame's content needs to be retrieved separately from the server. This can result in additional network traffic and potentially slower page load times, especially if the server's capacity or the network's bandwidth is limited.

It's important to consider the performance implications when using frames or any other technique that involves multiple requests to the server. Optimizing the number and size of requests, as well as implementing caching and other performance-enhancing techniques, can help mitigate the impact on the server's load.

Question 9Not yet
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What is the output of the following Java fragment code?

```
int[] list = new int[4];
int sum = 0;
for(int i = 0; i < list.length; i++)
{
    list[i] = i*3;
    sum += list[i];
}
System.out.print(sum);
```

- ☐ a. 6
- ☐ b. 3
- ☐ c. 36
- ☐ d. 18

Answer 18;

Initially , length of list[4] array is 4. It is given and the loop iterates from i=0 upto the length of array list minus 1 i.e. 3. Sum+=list[i] means sum=sum+list[i];

Then at i=0;

Sum=0;

List[0]=0*3=0;

Sum=0+0=0;

At i=1;

List[1]=1*3=3;

Sum=0+3=3;

At i=2;

List[2]=2*3=6;

Sum=6+3=9;

At i=3;

List[3]=3*3=9;

Sum=9+9=**18**;

At i=4;

The iteration of loop terminated and the last statement cout executed

Therefore the output of sum is equals to 18;

Question 10

Not yet answered

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Which one of the following is a security service enforced to protect disclosure of information, which can be stored in file or being on transmission, from unauthorized entities?

- ☐ a. Authentication
- ☐ b. Confidentiality
- ☐ c. Integrity
- ☐ d. Availability

Answer B. Confidentiality;

The security service enforced to protect the disclosure of information, whether stored in a file or being transmitted, from unauthorized entities is Confidentiality.

Confidentiality ensures that sensitive information remains private and accessible only to authorized individuals or entities. It prevents unauthorized disclosure, access, or interception of data. Measures such as encryption, access controls, and secure communication protocols are implemented to maintain the confidentiality of information.

Options A. Authentication, B. Integrity, and C. Availability are also important security services, but they do not specifically address the protection of information from unauthorized disclosure.

Authentication refers to the process of verifying the identity of a user or entity, ensuring that they are who they claim to be.

Integrity focuses on maintaining the accuracy and completeness of data, ensuring that it remains unaltered and intact during storage or transmission.

Availability pertains to the accessibility and reliability of information and systems, ensuring that they are accessible to authorized users when needed.

While all these security services are crucial, the specific service that protects against unauthorized disclosure of information is Confidentiality. Therefore, option D. Confidentiality is the correct answer.

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Question 11

Not yet answered

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A computer expert would like to know the number of processes that are completed per hour to assess the performance of a machine. Therefore, which one of the following best describes the expert's assessment?

- ☐ a. Waiting time
- ☐ b. System throughput
- ☐ c. Response time
- ☐ d. Turnaround time

B. system throughput

System throughput is typically expressed in terms of the number of tasks or operations completed per unit of time, such as tasks per second or operations per minute. A higher throughput indicates that the system is capable of processing a greater number of tasks, requests, or operations in a given time frame.

1. **Waiting Time:** Waiting time refers to the amount of time a process or task spends in the ready queue, waiting to be executed by the CPU. It includes the time a process waits for its turn to be scheduled by the operating system. Waiting time is an important metric as it directly affects the overall system performance and user experience. Minimizing waiting time is desirable to ensure efficient CPU utilization and timely execution of processes.
2. **Response Time:** Response time, also known as response latency, is the time elapsed between initiating a request or action and receiving the first response or output. It measures the speed or responsiveness of the system from the user's perspective. In the context of interactive systems, response time is critical, as it determines how quickly the system reacts to user input and provides feedback. Faster response times lead to a more interactive and user-friendly experience.
3. **Turnaround Time:** Turnaround time, also referred to as execution time or completion time, is the total time taken for a process or task to complete its execution from the moment it enters the system until it finishes. It includes the time spent in the ready queue, executing on the CPU, and waiting for I/O operations. Turnaround time is a comprehensive metric that reflects the overall performance of the system, considering both waiting and execution times. Minimizing turnaround time is crucial to optimize system efficiency and throughput.

These metrics are important considerations when designing and evaluating operating systems. Optimizing waiting time, response time, and turnaround time leads to improved system performance, better user experience, and efficient resource utilization. Operating system scheduling algorithms, I/O management, and process management techniques are employed to minimize these metrics and enhance system performance.

Question 12
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Which one of the following is correct way creating String object?

- ☐ a. `String m = 'H', 'e', 'l', 'l', 'o';`
- ☐ b. `String m = new String("Technology");`
- ☐ c. `String m = "Computer Science";`
- ☐ d. `String m = new String("Exit Exam");`

Answer D; it should be double quoted for string and also single quote is only to create character object using char keyword or literal;

```
String str = new String("Hello, World!"); //using new keyword
```

```
String str = "Hello, World!"; //using String literal
```

```
Char ch='A';
```

Question 13

Not yet answered

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Which one of the following computers can be most secured compared to the rest?

- ☐ a. A computer connected to the Internet having strong intrusion detection
- ☐ b. A computer that is not connected to the Internet
- ☐ c. A computer connected to the Internet with the latest firewall system
- ☐ d. A computer connected to the Internet and the latest anti-virus installed on it

Answer B;

Question 14

Not yet answered

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An interface that has multiple lines connecting input/output module and peripheral device and assures multiple bits to be transferred at the same time is:

- ☐ a. One line interface
- ☐ b. Bus interface
- ☐ c. Parallel interface
- ☐ d. Serial interface

Answer C;

In Computer Organization and Architecture (COA), parallel and serial interfaces are two methods of transmitting data between different components or devices. Here's an explanation of parallel and serial interfaces:

1. **Parallel Interface:** A parallel interface is a method of transferring multiple bits of data simultaneously using separate data lines. In a parallel interface, each bit of data is transmitted on its dedicated line. For example, an 8-bit parallel interface uses eight data lines to transfer eight bits of data simultaneously.

In a parallel interface, the data is typically transferred in parallel form from the source to the destination. This means that all bits of the data word are sent simultaneously on their respective lines. Parallel interfaces allow for high-speed data transfer as multiple bits can be transmitted simultaneously. They are commonly used in systems that require high data transfer rates, such as memory buses and internal communication within a computer system.

2. **Serial Interface:** A serial interface is a method of transmitting data sequentially, one bit at a time, over a single data line. In a serial interface, the bits of data are sent one after another in a continuous stream. The receiver at the other end of the serial interface reconstructs the original data by reassembling the received bits.

Serial interfaces are typically used in situations where the available data lines are limited, or when long-distance communication is required. Serial interfaces are more commonly used in external communication between devices, such as between a computer and a peripheral device (e.g., serial ports, USB, Ethernet). Serial communication allows for simpler and more cost-effective connections with fewer data lines.

Serial interfaces are slower compared to parallel interfaces since the data is transmitted sequentially, one bit at a time. However, advancements in serial communication protocols and technologies, such as higher baud rates and efficient encoding schemes, have made serial interfaces capable of achieving high data transfer rates.

In summary, parallel interfaces transfer multiple bits simultaneously over separate data lines, allowing for high-speed data transfer. Serial interfaces transmit data sequentially over a single data line, which is useful for communication over limited connections or long distances. The choice between parallel and serial interfaces depends on factors such as data transfer requirements, available resources, and the distance between communicating devices.

Question 15

Not yet answered

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Which network type is the largest as compared to the rest?

- ☐ a. Metropolitan Area Network
- ☐ b. The Internet
- ☐ c. Local Area Network
- ☐ d. Wide Area Network

The largest network type from personal to the internet is the Internet itself. Let's explain the coverage area of each network type:

1. **Personal Area Network (PAN):** A Personal Area Network (PAN) is the smallest network type, typically covering a very short range, often within a person's personal space. It connects devices such as smartphones, laptops, tablets, and peripherals like Bluetooth speakers or keyboards. PANs are designed for personal convenience and communication between devices in close proximity, typically within a few meters.
2. **Local Area Network (LAN):** A Local Area Network (LAN) covers a small geographical area such as a home, office building, or campus. LANs connect devices within the same physical location, enabling data sharing, resource sharing, and communication between connected devices. LANs are usually connected using Ethernet cables or Wi-Fi technology, allowing devices to communicate at high speeds within the local network.
3. **Metropolitan Area Network (MAN):** A Metropolitan Area Network (MAN) covers a larger geographical area, such as a city or town. MANs interconnect multiple LANs across a metropolitan area to facilitate communication and resource sharing between different organizations or institutions. MANs typically use fiber optic cables or other high-speed connections to provide fast data transfer rates over longer distances.
4. **Wide Area Network (WAN):** A Wide Area Network (WAN) covers a wide geographical area, often spanning across cities, countries, or even continents. WANs connect multiple LANs and MANs over long distances. The most notable example of a WAN is the Internet itself, which connects

networks and devices worldwide. WANs rely on various technologies, including leased lines, satellite links, and internet service providers (ISPs), to provide connectivity over large distances.

The Internet, being the largest network type, has a global coverage area, connecting billions of devices across the world. It enables communication, information sharing, and access to resources on a global scale.

Each network type plays a crucial role in providing connectivity and facilitating communication at different levels, from personal devices to global connectivity. They cater to different coverage areas, ranging from personal spaces (PANs) to small local environments (LANs), larger city or town regions (MANs), and finally, the extensive global reach of the Internet (WAN)

Question 16

Not yet answered

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Which one of the following is true about java and javascript?

- ☐ a. Javascript can be used in extremely sophisticated applications
- ☐ b. Javascript is compiled and interpreted in source code form
- ☐ c. Both are object oriented languages
- ☐ d. There are more features in javascript than java

Answer C;

Question 17

Not yet answered

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Which one of the following is not a single level ordered index?

- ☐ a. Secondary index
- ☐ b. Primary index
- ☐ c. Multilevel index
- ☐ d. Clustering index

To clarify, a multilevel index is not a single-level ordered index. It is an indexing technique that uses multiple levels or tiers of indexes to efficiently organize and access data records. Each level contains a subset of the keys from the previous level, forming a hierarchical structure. This hierarchical organization helps in reducing the search space and improving search performance, especially for large databases.

In contrast, the other options, A. secondary index, C. clustering index, and D. primary index, can all be considered as types of single-level ordered indexes.

Question 18

Not yet answered

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What is the time complexity order of Quick sort algorithm?

- ☐ a. $O(2^n)$
- ☐ b. $O(n \log_2 n)$
- ☐ c. $O(n^2)$
- ☐ d. $O(n)$

Answer C;

Here is a summary of the time and space complexity for common searching and sorting algorithms, including their best, worst, and average case scenarios, in terms of big O notation:

Searching Algorithms:

1. Linear Search:

- Time Complexity:
 - Best Case: $O(1)$ - when the target element is found at the beginning of the list.
 - Worst Case: $O(n)$ - when the target element is at the end of the list or not present.
 - Average Case: $O(n)$ - since the target element could be anywhere in the list on average.
- Space Complexity: $O(1)$ - constant space, as it does not require additional data structures.

2. Binary Search:

- Time Complexity:
 - Best Case: $O(1)$ - when the target element is found at the middle position.
 - Worst Case: $O(\log n)$ - when the target element is not present, and the search space is halved at each step.
 - Average Case: $O(\log n)$ - logarithmic time complexity due to the halving of the search space.
- Space Complexity: $O(1)$ - constant space, as it does not require additional data structures.

Sorting Algorithms:

1. Bubble Sort:

- Time Complexity:
 - Best Case: $O(n)$ - when the input array is already sorted.
 - Worst Case: $O(n^2)$ - when the input array is in reverse order.
 - Average Case: $O(n^2)$ - quadratic time complexity due to nested iterations.
- Space Complexity: $O(1)$ - constant space, as it performs in-place swapping.

2. Selection Sort:

- Time Complexity:

- Best Case: $O(n^2)$ - same as the worst and average cases.
- Worst Case: $O(n^2)$ - when the input array is in any order.
- Average Case: $O(n^2)$ - quadratic time complexity due to nested iterations.
- Space Complexity: $O(1)$ - constant space, as it performs in-place swapping.

3. Insertion Sort:

- Time Complexity:
 - Best Case: $O(n)$ - when the input array is already sorted.
 - Worst Case: $O(n^2)$ - when the input array is in reverse order.
 - Average Case: $O(n^2)$ - quadratic time complexity due to shifting elements.
- Space Complexity: $O(1)$ - constant space, as it performs in-place shifting.

4. Merge Sort:

- Time Complexity:
 - Best Case: $O(n \log n)$ - same as the worst and average cases.
 - Worst Case: $O(n \log n)$ - when the input array is in any order.
 - Average Case: $O(n \log n)$ - due to the divide-and-conquer strategy.
- Space Complexity: $O(n)$ - additional space is required for the temporary array during merging.

5. Quick Sort:

- Time Complexity:
 - Best Case: $O(n \log n)$ - when the pivot divides the array evenly.
 - Worst Case: $O(n^2)$ - when the pivot is the smallest or largest element.
 - Average Case: $O(n \log n)$ - expected time complexity due to the randomized partitioning.
- Space Complexity: $O(\log n)$ - due to the recursive nature of the algorithm.

Please note that these complexities represent general characteristics and may vary depending on specific implementations and variations of the algorithms.

6. Shell Sort:

- Time Complexity:

- Best Case: $O(n \log n)$ - for certain gap sequences.
- Worst Case: $O(n^2)$ - when the gap sequence is not optimal.
- Average Case: Depends on the gap sequence. It varies between $O(n \log n)$ and $O(n^2)$.
- Space Complexity: $O(1)$ - Shell Sort is an in-place sorting algorithm.

7. Heap Sort:

- Time Complexity:
 - Best Case: $O(n \log n)$ - when the input is already a valid max/min heap.
 - Worst Case: $O(n \log n)$ - regardless of the input order.
 - Average Case: $O(n \log n)$ - heap sort has the same time complexity in all cases.
- Space Complexity: $O(1)$ - Heap Sort is an in-place sorting algorithm.

Heap Sort has a better worst-case time complexity compared to Shell Sort, but it requires additional memory to build and maintain the heap structure during sorting. Shell Sort, on the other hand, is an in-place sorting algorithm but can have variable performance depending on the chosen gap sequence.

It's important to note that the time and space complexities mentioned here represent the general characteristics of these algorithms and may vary depending on the specific implementation and variations used.

Question 19
Not yet answered
Marked out of 1.00
[Flag question](#)

Which asymptotic notation describes worst case analysis?

- ☐ a. $f(n) = \text{little-}\omega$ (little omega)
- ☐ b. $f(n) = \theta$ (thetha)
- ☐ c. $f(n) = \Omega$ (Big omega)
- ☐ d. $f(n) = \text{little-oh}$

Little o asymptotic notation

Big-O is used as a tight upper bound on the growth of an algorithm's effort (this effort is described by the function $f(n)$), even though, as written, it can also be a loose upper bound. "Little-o" ($o()$) notation is used to describe an upper bound that cannot be tight.

Question 20Not yet
answeredMarked out of
1.00

Flag question

Assume you make modification to your software. Then you want to make sure that the modification has not had introduced new errors. Which software testing level is required?

- ☐ a. Beta testing
- ☐ b. Dynamic testing
- ☐ c. Regression testing
- ☐ d. Static testing

Regression testing is testing existing software applications to make sure that a change or addition hasn't broken any existing functionality. Alpha testing is the first phase of formal testing, during which the software is tested internally using white-box techniques. Beta testing is the next phase, in which the software is tested by a larger group of users, typically outside of the organization that developed it.

Static Testing	Dynamic Testing
It is performed in the early stage of software development	It is performed at the later stage of the software development.
Testing is done without executing the program	Testing is done by executing the program
Also known as Verification testing	Also known as validation testing
It consists of reviews, walkthroughs, inspection, etc.,	It consists of functional, and non-functional testing, and data/control flow analysis.
Static testing assesses code and documentation	Dynamic testing gives bugs/bottlenecks in the software system.
Performs a dry run on the code as part of the static analysis.	Code is fully analyzed for different paths by executing it.
It generally takes a shorter time.	It usually takes a longer time as it involves running several test cases.
Static testing covers structural and statement coverage testing.	Dynamic testing covers the executable file of the code.
It includes requirement documents, design documents, program specifications, etc.,	It includes unit testing, integration testing, system testing, performance testing, security testing, etc.,

Question 21

Not yet answered

Marked out of 1.00

Flag question

What is the output of the following fragment C++ code?

```
for(int n = 1; n <= 18; n = n + 2)
{
    if(n % 7 != 0)
        cout<<n<<" ";
    else
        break;
}
```

- ☐ a. 1 3 5 7 9 11 13 15 17
- ☐ b. 1 2 3 4 5 6
- ☐ c. 1 3 5
- ☐ d. 3 5 7

Inside the **main()** function, there is a for loop that initializes an integer variable **n** with a value of 1. The loop continues as long as **n** is less than or equal to 18. After each iteration, **n** is incremented by 2 (**n = n + 2**), ensuring that **n** remains an odd number.

Within the for loop, there is an if-else statement that checks whether **n** is divisible by 7 (**n % 7 != 0**). If the condition is true, the value of **n** is not a multiple of 7, and the code inside the **if** block is executed. In this case, the code prints the value of **n** followed by a space (**cout << n << " "**).

If the condition is false, indicating that **n** is a multiple of 7, the code inside the **else** block is executed. In this case, the **break** statement is encountered, which terminates the execution of the for loop.

As a result, the program outputs the odd numbers from 1 to 18 (excluding multiples of 7) separated by spaces. The output would be:

1 3 5

After the loop finishes execution, the program returns 0 to indicate successful execution and terminates.

Here's an explanation of the break, continue, and goto statements in C++ along with simple sample code for each:

1. **break**: The **break** statement is used to terminate the execution of a loop or switch statement. When encountered, it immediately exits the loop or switch, and control is passed to the next statement after the loop or switch.

Example:

```

#include <iostream>
using namespace std;

int main() {
    for (int i = 1; i <= 5; i++) {
        if (i == 3) {
            break; // Terminate the loop when i is equal to 3
        }
        cout << i << " ";
    }
    cout << "Loop terminated." << endl;
    return 0;
}

```

Output:

vbnet

 Copy code

```
1 2 Loop terminated.
```

In this example, the for loop runs from **i = 1** to **i = 5**. When **i** becomes equal to 3, the **break** statement is encountered, terminating the loop. As a result, only the numbers 1 and 2 are printed before the loop is terminated.

2. **continue**: The **continue** statement is used to skip the current iteration of a loop and continue with the next iteration. When encountered, it jumps to the next iteration, bypassing the remaining code within the loop block.

Example:

cppCopy code

```
#include <iostream>
using namespace std;

int main() {
    for (int i = 1; i <= 5; i++) {
        if (i == 3) {
            continue; // Skip the current iteration when i is equal to 3
        }
        cout << i << " ";
    }
    cout << "Loop completed." << endl;
    return 0;
}
```

Output:

vbnet

 Copy code

```
1 2 4 5 Loop completed.
```

In this example, the for loop runs from $i = 1$ to $i = 5$. When i becomes equal to 3, the **continue** statement is encountered. It skips the remaining code in the current iteration and proceeds to the next iteration. As a result, the number 3 is skipped, and the loop continues until completion.

3. **goto**: The **goto** statement is used to transfer the control of the program to a labeled statement within the same function. It provides an unconditional jump to the specified label.

Example:

```

#include <iostream>
using namespace std;

int main() {
    int num = 0;
    start:
    cout << "Enter a positive number: ";
    cin >> num;

    if (num <= 0) {
        cout << "Invalid number. Try again." << endl;
        goto start; // Jump back to the 'start' label
    }

    cout << "The number you entered is: " << num << endl;
    return 0;
}

```

Output:

yml

 Copy code

```

Enter a positive number: -5
Invalid number. Try again.
Enter a positive number: 10
The number you entered is: 10

```

In this example, the program prompts the user to enter a positive number. If the entered number is not positive (≤ 0), the **goto** statement is used to jump back to the **start** label, prompting the user to re-enter a valid number. This process continues until a positive number is entered.

Please note that the **goto** statement should be used sparingly and with caution, as excessive use can make the code difficult to understand and maintain.

Question 22Not yet
answeredMarked out of
1.00

Flag question

What is the output of the following C++ fragment code?

```
int a, b;  
a= 13;  
b = 9;  
while( (20 && 0) && (a > b) )  
{  
    cout<<"Plants are our life";  
}
```

- ☐ a. No message will be displayed
- ☐ b. It displays message "Plants are our life" infinitely
- ☐ c. It displays message "Plants are our life" 20 times
- ☐ d. It displays message "Plants are Our Life" 4 times

Inside the while loop condition, there are two sub-conditions: **(20 && 0)** and **(a > b)**.

1. **(20 && 0)** represents the logical AND operation between the constants 20 and 0. However, both 20 and 0 are treated as boolean values, **where any non-zero value is considered true and 0 is considered false**. In this case, the condition evaluates to **false** since 0 is considered false in boolean context.
2. **(a > b)** represents a comparison between the variables **a** and **b**. In the given code, **a** is assigned a value of 13, and **b** is assigned a value of 9. Since 13 is indeed greater than 9, this condition evaluates to **true**.

However, the overall while loop condition **(20 && 0) && (a > b)** evaluates to **false** because one of the sub-conditions **(20 && 0)** is false. Therefore, the code inside the while loop, which outputs "Plants are our life," is never executed.

As a result, the code finishes executing without producing any output on the console.

Computer Science

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Question 23

Not yet answered

Marked out of 1.00

Flag question

Assume you have a universe of Key U. Which one is false statement regarding Direct Address table and Hash Table data structures?

- ☐ a. If universe U is very large, Hash table is better than Direct Address table
- ☐ b. Hash table allocates one separate memory slot for each key in the universe U
- ☐ c. There is one to one correspondence between keys in the universe U and memory slots in the Direct Address table
- ☐ d. Direct Address table doesn't use hash function to map keys

Previous question

Next question

Answer B;

No, the statement that a hash table allocates one separate memory slot for each key in the Universe U is not true. In a typical hash table implementation, the number of memory slots or buckets in the hash table is generally much smaller than the size of the universe of keys, U.

The purpose of a hash table is to map keys from the universe U to a smaller range of memory slots using a hash function. The hash function determines the index or bucket where a key should be stored. Therefore, the number of memory slots in a hash table is typically determined by the capacity of the hash table, which is typically much smaller than the size of the universe U.

Choice C is true, the statement "there is a one-to-one correspondence between keys in the universe U and memory slots in the direct address table" is true for a direct address table.

In a direct address table, each key from the universe U is directly mapped to a specific memory slot or array index. The index in the array represents a unique key from the universe U.

For example, if the universe U consists of integers from 0 to 9, a direct address table with an array of size 10 would have a one-to-one correspondence between the keys in U and the memory slots in the array. The key "0" would be stored in index 0, the key "1" would be stored in index 1, and so on.

Comparison between the two is given below;

Hash tables and direct address tables are both data structures used to store and retrieve elements based on keys. However, they differ in terms of their implementation and characteristics. Let's compare and contrast hash tables and direct address tables in the context of a given universe of keys U:

Hash Table:

- **Implementation:** A hash table uses a hash function to map keys from the universe U to an array of buckets or slots.
- **Memory Usage:** Hash tables typically use less memory compared to direct address tables because they only allocate memory for the elements actually stored.
- **Collisions:** Due to the potential for hash collisions (when two different keys are mapped to the same hash value), hash tables use collision resolution techniques such as chaining or open addressing to handle multiple keys that hash to the same bucket.
- **Access Time:** In the average case, the time complexity for accessing an element in a hash table is $O(1)$. However, in the worst case (when there are many collisions), the time complexity can degrade to $O(n)$, where n is the number of elements stored.
- **Key Requirements:** The keys in a hash table must be hashable, meaning they must have a well-defined hash function that distributes the keys uniformly across the array of buckets.

Direct Address Table:

- **Implementation:** A direct address table uses an array of size $|U|$ (the size of the universe) directly indexed by the keys. Each index in the array corresponds to a specific key from the universe U .
- **Memory Usage:** Direct address tables consume more memory compared to hash tables because they allocate memory for each possible key in the universe, regardless of whether elements are actually stored or not.
- **Collisions:** Direct address tables do not suffer from collisions since each key is mapped directly to an array index. Each index represents a unique key.
- **Access Time:** Accessing an element in a direct address table is typically a constant time operation ($O(1)$) since the index is known directly based on the key. However, this assumes that the size of the universe is manageable and that memory is available to allocate an array of that size.
- **Key Requirements:** Direct address tables require that the universe U of keys is small and enumerable. Each possible key must be unique and have a one-to-one mapping with an array index.

In summary, the choice between a hash table and a direct address table depends on factors such as the size of the universe of keys, memory constraints, and the expected number of elements to be stored. Hash tables are more flexible, handle collisions, and have an average constant time complexity for access. Direct address tables offer constant time access but require significant memory for large universes.

Computer Science

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Question 24

Not yet answered

Marked out of 1.00

Flag question

A process is terminated by the operating system due to a reference to non-existing memory. Which one of the following best explains the process termination?

- ☐ a. Normal exit
- ☐ b. Terminated by another process
- ☐ c. Error exit
- ☐ d. Fatal exit

Previous question

Next question

Answer D;

In operating systems, process termination can occur in various ways, depending on the circumstances. Here are definitions and examples of different types of process termination:

1. **Normal Exit:** A process terminates normally when it completes its execution successfully and exits voluntarily. This typically happens when the process reaches the end of its main function or explicitly calls an exit system call.

Example: Consider a program that reads a file, performs some computations, and then writes the result to another file. If the program completes the computation and writing process without any errors, it terminates normally.

2. **Error Exit:** An error exit occurs when a process encounters an error condition during its execution and terminates as a result. This can happen due to various reasons such as invalid input, resource unavailability, or internal errors within the process.

Example: Suppose a program is designed to divide two numbers entered by the user. If the user enters zero as the divisor, the program detects the error and terminates with an error message indicating division by zero.

3. **Fatal Exit:** A fatal exit happens when a process encounters a critical error or unrecoverable condition, leading to its termination. This could be due to severe hardware failures, memory corruption, or other catastrophic events that make it impossible for the process to continue executing.

Example: If a program attempts to access memory outside of its allocated range or encounters a hardware fault that cannot be handled, the operating system may forcibly terminate the process to prevent further damage.

4. **Terminated by Another Process:** A process can be terminated by another process or by the operating system itself. This can occur due to various reasons, such as resource management, policy enforcement, or user intervention. The terminating process or the operating system typically sends a termination signal to the target process, indicating that it should stop executing.

Example: A user running multiple processes in a command-line interface may choose to manually terminate a process by sending an appropriate termination signal, such as pressing Ctrl+C, to interrupt the execution and terminate the process.

These are some common types of process termination in an operating system. Each type has its own characteristics and reasons for termination, which can vary depending on the specific circumstances and context of the executing process

Computer Science

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Question 25

Not yet answered

Marked out of 1.00

Flag question

Which phase of software development process refers to modification of software to reflect changing customer and market requirements?

- ☐ a. Specification
- ☐ b. Evolution
- ☐ c. Development
- ☐ d. Validation

Previous question

Next question

D. Evolution

Computer Science

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Time left 1:01:13

Question 26

Not yet
answered

Marked out of
1.00

Flag question

Which one of the following is not correct about POP3 and SMTP?

- ☐ a. A sender can send email using SMTP
- ☐ b. Both are application layer protocols
- ☐ c. Both are network layer protocols
- ☐ d. A sender can receive emails using POP3

Previous question

Next question

Answer C;

Computer Science

Back

Time left 1:01:07

Question 27

Not yet
answered

Marked out of
1.00

Flag question

Which one of the following is correct for the statement below? "A problem that occurs when one transaction updates a database item and then the transaction fails for some reason and the updated item is accessed by another transaction before it is changed back to its original value"

- ☐ a. It is an incorrect summary problem
- ☐ b. It is a lost update problem
- ☐ c. It is dirty read
- ☐ d. It is a temporary read problem

Previous question

Next question

Answer C;

Computer Science

Back

Time left 1:00:59

Question 28

Not yet answered

Marked out of 1.00

Flag question

Given the schemas:BOOK (Book_id, Title, Publisher_name) BOOK_AUTHOR (Book_id, Author_name) PUBLISHER (Name, Address, Phone) Which one of the following SQL is correct to delete the publisher named as 'XWZ' from the PUBLISHER table

- ☐ a. DELETE FROM PUBLISHER WHERE Address = 'XWZ'
- ☐ b. DELETE FROM PUBLISHER WHERE Name = 'XWZ'
- ☐ c. DELETE FROM BOOK_AUTHOR WHERE Address = 'XWZ'
- ☐ d. DELETE FROM BOOK WHERE Name = 'XWZ'

Previous question

Next question

Answer B;

Computer Science

Back

Time left 1:00:51

Question 29

Not yet answered

Marked out of 1.00

Flag question

Which one of the following is responsible for separating some machines from the rest to shield them from possible attacks?

- ☐ a. Network operating system
- ☐ b. Router
- ☐ c. Firewall
- ☐ d. Switch

Previous question

Next question

Answer C. Firewall

Computer Science

Back

Time left 1:00:45

Question 30

Not yet answered

Marked out of 1.00

Flag question

Assume you have a weighted graph G with a set of vertices V and set of edges E . Which one of the following problem is an optimization problem on the graph?

- ☐ a. How many cycles are there in graph G
- ☐ b. Traverse all vertices in graph G
- ☐ c. Find minimum spanning tree from graph G
- ☐ d. Is vertex $p \in V$ reachable from vertex $q \in V$ in graph G

Previous question

Next question

C. finding minimum spanning tree

Computer Science

Back

Time left 1:00:40

Question 31

Not yet answered

Marked out of 1.00

Flag question

Which one of the following lists of environments is the hardest for an agent?

- ☐ a. Fully observable, sequential, deterministic
- ☐ b. Sequential, non-deterministic, dynamic
- ☐ c. Deterministic, fully observable, static
- ☐ d. Partially observable, static, continuous

Previous question

Next question

Answer B;

Computer Science

Back

Time left 1:00:35

Question 32

Not yet answered

Marked out of 1.00

Flag question

Which one of the following is used for the purpose of syntax analysis?

- ☐ a. Scanner
- ☐ b. Parser
- ☐ c. Loader
- ☐ d. Linker

Previous question

Next question

B. Parser

Computer Science

Back

Time left 1:00:31

Question 33

Not yet answered

Marked out of 1.00

Flag question

Which one of the following is different from the other based on its role in computer network?

- ☐ a. Hubs
- ☐ b. Routers
- ☐ c. Network printer
- ☐ d. Wireless access point

Previous question

Next question

C. Network printer

Computer Science

Back

Time left 1:00:26

Question 34

Not yet answered

Marked out of 1.00

Flag question

In a tree data structure, if a node has no parent node, then the node is _____.

- ☐ a. Root node
- ☐ b. Parent node
- ☐ c. External node
- ☐ d. Internal node

Previous question

Next question

A. Root node

Computer Science

Back

Time left 1:00:22

Question 35

Not yet answered

Marked out of 1.00

Flag question

Which phase of compiling process is considered as the final phase in the process?

- ☐ a. Intermediate code generation
- ☐ b. Code optimization
- ☐ c. Semantic analysis
- ☐ d. Code generation

Previous question

Next question

D. code generation

Computer Science

Back

Time left 1:00:19

Question 36

Not yet answered

Marked out of 1.00

🚩 Flag question

Which one of the following does not show non-functional requirement of software systems?

- ☐ a. Reliability
- ☐ b. Response time
- ☐ c. Displaying information
- ☐ d. Memory requirement

Previous question

Next question

C Display information

Computer Science

Back

Time left 1:00:13

Question 37

Not yet answered

Marked out of 1.00

🚩 Flag question

Which deadlock prevention mechanisms don't require timestamp?

- ☐ a. Wound-wait
- ☐ b. Wait-wait
- ☐ c. No-wait
- ☐ d. Wait-die

Previous question

Next question

C No wait

Computer Science

Back

Time left 1:00:09

Question 38

Not yet answered

Marked out of 1.00

Flag question

What does the program data independence in the database approach entails for?

- ☐ a. Data and applications are defined separately
- ☐ b. Both program and data are defined together
- ☐ c. The application may be affected when data changes
- ☐ d. Programs can create redundant data independently

Previous question

Next question

Answer A;

Computer Science

Back

Time left 1:00:05

Question 39

Not yet answered

Marked out of 1.00

Flag question

Which one of the following is not correct about computer architecture and organization?

- ☐ a. Manufacturers offer computer with same architecture but different organization
- ☐ b. An architecture can be used for long time encompassing different computer models
- ☐ c. An organization can last for long time as an architecture does
- ☐ d. An architecture can remain for long time but its organization can change

Previous question

Next question

Answer B;

Computer Science

Back

Time left 0:59:58

Question 40

Not yet answered

Marked out of 1.00

Flag question

Given input Alphabet $\Sigma = \{a, b, c, d\}$ and empty string λ . Then, which one of the following is equivalent to Σ^0 ??

- ☐ a. Σ
- ☐ b. $\{\lambda\}$
- ☐ c. Σ^*
- ☐ d. Σ^+

Previous question

Next question

Answer B;

Computer Science

Back

Time left 0:59:50

Question 41

Not yet answered

Marked out of 1.00

Flag question

Suppose you are administering a network and would like to know whether a computer in the upstairs is connected to the network or not. Therefore, which command would you use?

- ☐ a. ping
- ☐ b. traceroute
- ☐ c. DHCP discover
- ☐ d. nslookup

Previous question

Next question

Answer A. Ping

Computer Science

Back

Time left 0:59:45

Question 42

Not yet answered

Marked out of 1.00

Flag question

What is the output of the following Java fragment code?

```
int n = 6, m = 15, p = 3;  
n -= m;  
n *= p;  
System.out.print(n);
```

- ☐ a. 27
- ☐ b. -27
- ☐ c. 3
- ☐ d. 24

Previous question

Next question

Answer B. -27;

Computer Science

Back

Time left 0:59:31

Question 43

Not yet answered

Marked out of 1.00

Flag question

Which one is a method used to obtain the meaning of encrypted information without access to secret key?

- ☐ a. Encryption
- ☐ b. Cryptography
- ☐ c. Decryption
- ☐ d. Crypto-analysis

Previous question

Next question

D. crypto-analysis

Computer Science

Back

Time left 0:59:06

Question 44

Not yet answered

Marked out of 1.00

Flag question

Which one of the following is a two-level cache having an internal and external caches?

- ☐ a. Single level cache
- ☐ b. Unified cache
- ☐ c. Multilevel cache
- ☐ d. Split cache

Previous question

Next question

Ans D split catch

Computer Science

Back

Time left 0:58:59

Question 45

Not yet answered

Marked out of 1.00

Flag question

Advancement of operating system is closely tied to computer architectures, therefore, has passed through different generations. Which one of the following generation and advancement is wrongly coupled?

- ☐ a. Third generation -> batch system
- ☐ b. Second generation -> multiprogramming
- ☐ c. First generation -> single user
- ☐ d. Fourth generation -> real time systems

Previous question

Next question

Ans: B

Computer Science

Back

Time left 0:58:55

Question 46

Not yet
answered

Marked out of
1.00

Flag question

Suppose you have received a message from your friend but the message is scrambled for the sake of security. Your friend has used a program to encode messages into scrambled text. In order to read the message which program would you use?

- ☐ a. Decryption program
- ☐ b. Secret key
- ☐ c. Encryption program
- ☐ d. Ciphertext

Previous question

Next question

Ans: A

Computer Science

Back

Time left 0:58:51

Question 47

Not yet
answered

Marked out of
1.00

Flag question

Look at the following PHP line: `$list = array(20, 12, 23, 19, 18);` What is the index of data value 19?

- ☐ a. 1
- ☐ b. 0
- ☐ c. 3
- ☐ d. 2

Previous question

Next question

Ans: C

Computer Science

Back

Time left 0:58:46

Question 48

Not yet answered

Marked out of 1.00

Flag question

After deep analysis if your software requires major change to the architecture of the system, which strategy of software change should be applied?

- ☐ a. Preventative maintenance
- ☐ b. Software maintenance
- ☐ c. Software reengineering
- ☐ d. Architectural transformation

Previous question

Next question

Ans: D

Computer Science

Back

Time left 0:58:35

Question 49

Not yet answered

Marked out of 1.00

Flag question

_____ is a network architecture characteristics that states a network can expand quickly to address needs of new users and application without having impact on the services delivered to existing users.

- ☐ a. Scalability
- ☐ b. Fault tolerance
- ☐ c. Quality of service
- ☐ d. Security

Previous question

Next question

Ans: A;

Computer Science

Back

Time left 0:58:30

Question 50

Not yet answered

Marked out of 1.00

Flag question

At which layer of the OSI model devices such as bridges, switches and Network Interface cards are used?

- ☐ a. Network layer
- ☐ b. Data link layer
- ☐ c. Application layer
- ☐ d. Physical layer

Previous question

Next question

B Data link layer

Computer Science

Back

Time left 0:58:26

Question 51

Not yet answered

Marked out of 1.00

Flag question

Which type of variable can be accessed throughout our C++ program scope?

- ☐ a. Static
- ☐ b. Automatic
- ☐ c. Local
- ☐ d. Global

Previous question

Next question

D Global

Computer Science

Back

Time left 0:58:19

Question 52

Not yet answered

Marked out of 1.00

Flag question

Which one of the following is correct for the statement below? When two transactions that access the same database items have their operations interleaved in a way that makes the value of some database item incorrect?

- ☐ a. It is a temporary read problem
- ☐ b. It is a dirty read problem
- ☐ c. It is a lost update problem
- ☐ d. It is an incorrect summary problem

Previous question

Next question

C- lost update or write write conflict

Computer Science

Back

Time left 0:58:14

Question 53

Not yet answered

Marked out of 1.00

Flag question

If you want to allow subclasses to access data fields or methods defined in the super-class, but not to allow non-subclasses to access these data fields and methods, which access modifier is appropriate?

- ☐ a. public
- ☐ b. default
- ☐ c. protected
- ☐ d. private

Previous question

Next question

Answ; C protected

Computer Science

Back

Time left 0:58:07

Question 54

Not yet answered

Marked out of 1.00

Flag question

Which layer uses port number to identify applications?

- ☐ a. Network layer
- ☐ b. Transport layer
- ☐ c. Physical layer
- ☐ d. Application layer

Previous question

Next question

Asw: B transport layer

Computer Science

Back

Time left 0:58:00

Question 55

Not yet answered

Marked out of 1.00

Flag question

Assume there is a line of patients waiting for a service from a clinic. Seriously sick patients should get service first. Which data structure is the most appropriate for the given type of real world problem?

- ☐ a. Priority Queue
- ☐ b. Queue
- ☐ c. Stack
- ☐ d. Tree

Previous question

Next question

Ans Priority queue

Computer Science

Back

Time left 0:57:55

Question 56

Not yet
answered

Marked out of
1.00

Flag question

Which statement is false about universal hashing technique?

- ☐ a. In universal hashing a hash function is selected randomly from set of hash functions for each key
- ☐ b. Single fixed hash function technique is better than universal hashing technique in terms of collision reduction
- ☐ c. In universal hashing a hash function is selected from set of hash functions independent of keys.
- ☐ d. Universal hashing reduces the chance of data collision

Previous question

Next question

Ans: A

Computer Science

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Time left 0:57:48

Question 57

Not yet
answered

Marked out of
1.00

Flag question

Which one of the following is not delimiter of PHP code?

- ☐ a. <caption> </caption>
- ☐ b. <? ?>
- ☐ c. <script language="PHP"> </script>
- ☐ d. <?php ?>

Previous question

Next question

Answer A;

Computer Science

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Time left 0:57:44

Question 58

Not yet answered

Marked out of 1.00

Flag question

Which one of the following is not correct about packet and circuit switching?

- ☐ a. In circuit switching a channel is dedicatedly used
- ☐ b. In Packet switching and end-to-end connection has to be established
- ☐ c. In packets switching messages are sent in small blocks
- ☐ d. Packet switching is more efficient than circuit switching

Previous question

Next question

Answer B;

Computer Science

Back

Time left 0:57:40

Question 59

Not yet answered

Marked out of 1.00

Flag question

What will be the output of the following JavaScript statement? `Math.floor(5.9)`

- ☐ a. 11.8
- ☐ b. 6
- ☐ c. 9
- ☐ d. 5

Previous question

Next question

Answer D;

Computer Science

Back

Time left 0:57:35

Question 60

Not yet answered

Marked out of 1.00

[Flag question](#)

Which one of the following is not correct for the figure shown in Question 27?

- ☐ a. P1 is waiting for a resource held by P2
- ☐ b. P0 is waiting for a resource held by P1
- ☐ c. P2 cannot be executed until P1 finishes
- ☐ d. P3 cannot be executed until P4 finishes

Previous question

Next question

Computer Science

Back

Time left 0:57:30

Question 61

Not yet answered

Marked out of 1.00

[Flag question](#)

In which situation that using database management system (DBMS) becomes costly and not recommended to use it?

- ☐ a. In embedded systems having too small storage space
- ☐ b. In situation where more than one users should access the database
- ☐ c. To deny access to unauthorized users
- ☐ d. To control redundancy

Previous question

Next question

Answer A;

Computer Science

Back

Time left 0:57:26

Question 62

Not yet answered

Marked out of 1.00

Flag question

58. Which network security issue shows the level of weakness in networks and devices?

- ☐ a. Treat
- ☐ b. Vulnerability
- ☐ c. Attack
- ☐ d. Availability

Previous question

Next question

Answer B vulnerability;

Computer Science

Back

Time left 0:57:21

Question 63

Not yet answered

Marked out of 1.00

Flag question

_____ is the whole thing that an agent has observed so far concerning the existing scenario in the environment.

- ☐ a. Perception
- ☐ b. Knowledge
- ☐ c. Action
- ☐ d. Performance measure

Previous question

Next question

Answer A;

Computer Science

Back

Time left 0:57:15

Question 64

Not yet answered

Marked out of 1.00

Flag question

Which statement is true about pop() operation of Stack data structures?

- ☐ a. pop() returns the end/top element but it doesn't delete that element
- ☐ b. pop() inserts new element at the middle of stack
- ☐ c. pop() removes the end/top element from stack
- ☐ d. pop() inserts new element at the beginning/bottom of stack

Previous question

Next question

Answer C;

Computer Science

Back

Time left 0:57:10

Question 65

Not yet answered

Marked out of 1.00

Flag question

Which one of the following is not an example of scripting languages?

- ☐ a. Perl
- ☐ b. Python
- ☐ c. C++
- ☐ d. PHP

Previous question

Next question

Answer C-C++

Computer Science

Back

Time left 0:57:05

Question 66

Not yet answered

Marked out of 1.00

Flag question

Which one of the following best explains attackers who are motivated by financial gain and perform sabotage, an IP theft and secrets of a rival company?

- ☐ a. Hacktivist
- ☐ b. Competitor
- ☐ c. Stealer
- ☐ d. Activist

Previous question

Next question

Ans: B;

Computer Science

Back

Time left 0:56:58

Question 67

Not yet answered

Marked out of 1.00

Flag question

For an input graph, Depth First Traversal algorithm uses _____ data structure to traverse.

- ☐ a. Queue
- ☐ b. Stack
- ☐ c. Priority queue
- ☐ d. Tree

Previous question

Next question

Answer C. stack

Question 68

Not yet
answered

Marked out of
1.00

Flag question

Let a Grammar $G = (\{S\}, \{a, b\}, S, P)$, where $\{S\}$ is set of variables, $\{a, b\}$ is set of terminal symbols, S is start variable and P is set of productions. Then which one of the following is not an element of the language (L) which is generated from G . Given:

$P : S \rightarrow aSb,$

$S \rightarrow \lambda$

- ☐ a. ab
- ☐ b. λ
- ☐ c. a
- ☐ d. $aabb$

Answer C;

Computer Science

Back

Time left 0:56:40

Question 69

Not yet
answered

Marked out of
1.00

Flag question

Which one of the following is not correct about transaction failure?

- ☐ a. Transactions fail if disk blocks lose their data
- ☐ b. Concurrency control enforcement avoids transaction failure
- ☐ c. Transactions fail if logical errors are detected
- ☐ d. Catastrophes can be cause of transaction failure

Previous question

Next question

Answer C;

Computer Science

Back

Time left 0:56:34

Question 70

Not yet answered

Marked out of 1.00

Flag question

Consider the following fragment of C++ code:

```
const int x = 8;
const int y = 16;
if(x > y)
{
    int z = x + y;
}
```

For codes like the above one, which technique of code optimization is appropriate?

- ☐ a. Dead code elimination
- ☐ b. Compile time evaluation
- ☐ c. Code motion
- ☐ d. Arithmetic simplification

Previous question

Next question

Answer : D;

Computer Science

Back

Time left 0:56:28

Question 71

Not yet answered

Marked out of 1.00

Flag question

Which one of the following is true about functions in JavaScript?

- ☐ a. Function names can begin with digits
- ☐ b. Function names are not case-sensitive
- ☐ c. Function names can contain spaces
- ☐ d. There is no limit to the number of function parameters that a function may contain

Previous question

Next question

Answer D;

Computer Science

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Time left 0:56:22

Question 72

Not yet answered

Marked out of 1.00

[Flag question](#)

If you plan a software test to be done by clients to demonstrate that the system satisfies their requirements, which phase of software testing is appropriate?

- ☐ a. System testing
- ☐ b. Acceptance testing
- ☐ c. Unit testing
- ☐ d. Integration testing

[Previous question](#)[Next question](#)

Answer B;

Computer Science

[Back](#)

Time left 0:56:18

Question 73

Not yet answered

Marked out of 1.00

[Flag question](#)

A database management system (DBMS) has _____ to control locks.

- ☐ a. Query processor
- ☐ b. Lock manager
- ☐ c. Query optimizer
- ☐ d. Lock table

[Previous question](#)[Next question](#)

Computer Science

Back

Time left 0:56:11

Question 74

Not yet answered

Marked out of 1.00

Flag question

Which asymptotic notation describes average case analysis?

- ☐ a. $f(n) = \theta$ (thetha)
- ☐ b. $f(n) = \Omega$ (Big omega)
- ☐ c. $f(n) = \text{little-}\omega$ (little omega)
- ☐ d. $f(n) = \text{Big-Oh}$

Previous question

Next question

Answer A;

Computer Science

Back

Time left 0:55:48

Question 75

Not yet answered

Marked out of 1.00

Flag question

Which one of the following is valid statement regarding PHP variables?

- ☐ a. `$test = 9.5`
- ☐ b. `$test = "Computer;`
- ☐ c. `$test = "This is 'Computer science' ";`
- ☐ d. `test = 5;`

Previous question

Next question

Answer C;

Computer Science

Back

Time left 0:55:41

Question 76

Not yet answered

Marked out of 1.00

Flag question

Which statement is false about programming language generations?

- ☐ a. Fifth generation programming languages need very powerful hardware and software
- ☐ b. Machine language can be executed without translation
- ☐ c. Assembly language is machine independent
- ☐ d. Assembly language should be translated to machine language before execution

Previous question

Next question

Ans: A; the most are platform independent

Computer Science

Back

Time left 0:55:36

Question 77

Not yet answered

Marked out of 1.00

Flag question

Which type of Turing machine has two tapes, one tape read-only and the other read-write tape?

- ☐ a. Multi-head Turing machine
- ☐ b. Non-deterministic Turing machine
- ☐ c. Multi-dimensional Turing machine
- ☐ d. Off-line Turing machine

Previous question

Next question

Answer B;

Computer Science

Back

Time left 0:55:29

Question 78

Not yet answered

Marked out of 1.00

Flag question

Among phases of compiling process, in which phase a sequence of characters are converted into a sequence of tokens?

- ☐ a. Syntax analysis
- ☐ b. Code optimization
- ☐ c. Lexical analysis
- ☐ d. Semantic analysis

Previous question

Next question

Answer C;

Computer Science

Back

Time left 0:55:15

Question 79

Not yet answered

Marked out of 1.00

Flag question

Which one of the following is false about String in Java?

- ☐ a. In Java a string is treated as object
- ☐ b. Content of a string can be changed once the string is created
- ☐ c. We can create a String object using String literal
- ☐ d. We can create a String object using array of characters

Previous question

Next question

Answer B; string is immutable(canot be changed) in java

Computer Science

Back

Time left 0:55:09

Question 80

Not yet answered

Marked out of 1.00

🚩 Flag question

Linked list data structure is _____ type of data structure.

- ☐ a. Static data structure
- ☐ b. Fixed size data structure
- ☐ c. Non-linear data structure
- ☐ d. Dynamic data structure

Previous question

Next question

Ans. D;

Computer Science

Back

Time left 0:55:04

Question 81

Not yet answered

Marked out of 1.00

🚩 Flag question

Which one of the following is evaluating the degrees of success of an agent?

- ☐ a. Performance measure
- ☐ b. Knowledge
- ☐ c. Action
- ☐ d. Perception

Previous question

Next question

Ans. A;

Computer Science

Back

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Question 82

Not yet answered

Marked out of 1.00

Flag question

Suppose you have developed a self-driving car equipped with relevant sensors and camera that helps it to see the road and drive sensing traffic lights, pedestrians and signs. Its sensors are able to detect the environment but not all the aspects. Hence, in which environment is your car acting?

- ☐ a. Stochastic
- ☐ b. Fully observable
- ☐ c. Partially observable
- ☐ d. Discrete

Previous question

Next question

Ans; Partially observable

Computer Science

Back

Time left 0:54:49

Question 83

Not yet answered

Marked out of 1.00

Flag question

Which one of the following phases is done before semantic analysis phase of compiling process?

- ☐ a. Code generation
- ☐ b. Syntax analysis
- ☐ c. Code optimization
- ☐ d. Intermediate code generation

Previous question

Next question

Answer B;

Computer Science

Back

Time left 0:54:43

Question 84

Not yet answered

Marked out of 1.00

Flag question

Which one of the following is a step in problem solving that lists the actions and states that the agent has to consider provided a goal?

- ☐ a. Execute
- ☐ b. Goal formulation
- ☐ c. Search
- ☐ d. Problem formulation

Previous question

Next question

Ans- problem formulation

Computer Science

Back

Time left 0:54:37

Question 85

Not yet answered

Marked out of 1.00

Flag question

Which quantifier can be used for some portion of the universe?

- ☐ a. $\rightarrow$
- ☐ b. $\forall$
- ☐ c. $\exists$
- ☐ d. $\wedge$

Previous question

Next question

Answer B;

Computer Science

Back

Time left 0:54:31

Question 86

Not yet answered

Marked out of 1.00

Flag question

Which one of the following algorithms can be used to extract Minimum Spanning Tree from an input graph?

- ☐ a. Merge sort algorithm
- ☐ b. Dijkstra's algorithm
- ☐ c. Prim's algorithm
- ☐ d. Huffman encoding algorithm

Previous question

Next question

Ans→Prim's algorithm

Computer Science

Back

Time left 0:54:25

Question 87

Not yet answered

Marked out of 1.00

Flag question

Which one of the following is false about project planning stage?

- ☐ a. Risk analysis is done during project planning
- ☐ b. Cost estimation is done during project planning
- ☐ c. Preparing time schedule is done during project planning
- ☐ d. Project planning is a one-time task in software development life cycle

Previous question

Next question

Answer→ A

Computer Science

[Back](#)

Time left 0:54:18

Question 88

Not yet
answered

Marked out of
1.00

🚩 Flag question

A university owns a number of private local area and wide area networks, which are designed for an access by its academic staff, students and administrative workers. Which term best describes the University's network?

- ☐ a. Wide Area Network
- ☐ b. The Internet
- ☐ c. Local Area Network
- ☐ d. Metropolitan Area Network

[Previous question](#)[Next question](#)

Answer B;

Computer Science

[Back](#)

Time left 0:54:11

Question 89

Not yet
answered

Marked out of
1.00

🚩 Flag question

Which one of the following is not correct about threads?

- ☐ a. It is a group of processes
- ☐ b. Treads have program counters
- ☐ c. Treads have registers to hold its working memory
- ☐ d. Treads are the entities schedule for execution on the CPU

[Previous question](#)[Next question](#)

Answer A;

Computer Science

Back

Time left 0:54:06

Question 90

Not yet answered

Marked out of 1.00

Flag question

What is the output of the following C++ fragment code?

```
int a = 6, b = 8;
int x = 2, y = 4;
int c = (x > y ? (a--, x) : (b--, y));
cout<<"a= "<<a;
cout<<"    b= "<<b;
cout<<"    c= "<<c;
```

- ☐ a. a = 6 b = 8 c = 4
- ☐ b. a = 5 b = 8 c = 4
- ☐ c. a = 6 b = 7 c = 4
- ☐ d. a = 5 b = 7 c = 2

Previous question

Next question

Here's the equivalent if-else statement for the given ternary operator expression:

```
#include <iostream>
using namespace std;

int main()
{
    int a = 6, b = 8, x = 2, y = 4;
    int c;

    if (x > y) {
        a--;
        c = x;
    } else {
        b--;
        c = y;
    }

    cout << "a: " << a << ", b: " << b << ", c: " << c << endl;

    return 0;
}
```

Answer C; 6 7 4

Computer Science

Back

Time left 0:54:01

Question 91

Not yet answered

Marked out of 1.00

Flag question

Which one of the following is a measure taken, to ensure information and information systems' availability, integrity, authentication and confidentiality?

- ☐ a. Wiretapping
- ☐ b. Interception
- ☐ c. Information assurance
- ☐ d. Information security

Previous question

Next question

Answer C information assurance

Question 92

Not yet answered

Marked out of 1.00

Flag question

Given input Alphabet $\Sigma = \{a, b, c, d\}$ and empty string λ . Then, which one of the following is true about Σ^* ?

- ☐ a. $\Sigma^* = \Sigma^1 \cup \Sigma^2 \cup \Sigma^3 \cup \Sigma^4$
- ☐ b. $\Sigma^* = \Sigma^+ \cup \{\lambda\}$
- ☐ c. $\Sigma^* = \{\lambda\}$
- ☐ d. $\Sigma^* = \Sigma^+$

Answer B;

Computer Science

Back

Time left 0:53:45

Question 93

Not yet answered

Marked out of 1.00

Flag question

Which one of data updates best describes the following statement? "All modified data items in the cache is written either after transaction ends its execution or after a pre-determined number of transactions have concluded their execution"

- ☐ a. Shadow update
- ☐ b. Immediate update
- ☐ c. Deferred update
- ☐ d. In-place update

Previous question

Next question

Answer C. Deferred

Computer Science

Back

Time left 0:53:38

Question 94

Not yet answered

Marked out of 1.00

Flag question

What is the purpose of an opcode that is as indicated in the figure below?



- ☐ a. It holds the source address of an operand
- ☐ b. It determines the operation to be done
- ☐ c. It holds the next instruction reference
- ☐ d. It holds the destination address of an operand

Previous question

Next question

Computer Science

Back

Time left 0:53:32

Question 95

Not yet answered

Marked out of 1.00

Flag question

Among the following, which one best describes a database management system (DBMS)?

- ☐ a. It can assist process images
- ☐ b. It is responsible for creating and maintaining a database
- ☐ c. It can assist you to process document
- ☐ d. It can help users to manage computer's activity

Previous question

Next question

Answer B;

Computer Science

Back

Time left 0:53:19

Question 96

Not yet answered

Marked out of 1.00

Flag question

Which one of the following is the smallest heading tag?

- ☐ a. <H1>
- ☐ b. <H6>
- ☐ c. <H4>
- ☐ d. <H3>

Previous question

Next question

Answer B;

Computer Science

Back

Time left 0:53:14

Question 97

Not yet answered

Marked out of 1.00

Flag question

What will be the output of the following JavaScript statement? `Math.sqrt(36)`

- ☐ a. 1
- ☐ b. 6
- ☐ c. 30
- ☐ d. 36

Previous question

Next question

Answer B;

Computer Science

Back

Time left 0:53:09

Question 98

Not yet
answered

Marked out of
1.00

Flag question

Which one of the following is false about arrays in C++?

- ☐ a. We can access elements of arrays without using index number
- ☐ b. An array is a collection of similar data objects
- ☐ c. Size of an array should be constant at the time of array declaration
- ☐ d. We use [] square bracket at the time of array declaration

Previous question

Next question

Answer A;

Computer Science

Back

Time left 0:53:03

Question 99

Not yet
answered

Marked out of
1.00

Flag question

Which branch of study deals with whether a problem can be solved at all or not, regardless of the resources required?

- ☐ a. Computability theory
- ☐ b. Complexity theory
- ☐ c. Automata theory
- ☐ d. Set theory

Previous question

Next question

Ans: C;

Computer Science

Back

Time left 0:52:57

Question 100

Not yet
answered

Marked out of
1.00

🚩 Flag question

Which one of the following is correct about human role in system administration?

- ☐ a. It doesn't need organizational skills
- ☐ b. System administrator is not expected to have skills and confidence
- ☐ c. Ethics is not a requirement to manage computers
- ☐ d. It requires patience, understanding and knowledge

Previous question

Finish exam ...

Answer D;